CE F230 Civil Engineering Materials

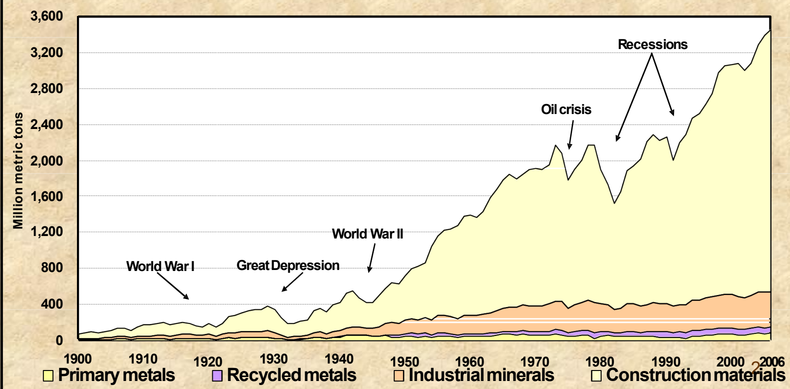
Dr. Mukund Lahoti, Instructor-in-charge

BITS Pilani
Pilani Campus

Common Materials used in Civil Engineering

Raw Materials Consumption

U.S. Raw, Nonfuel Mineral Materials Put into Use Annually, from 1900 through 2006
(Materials embedded in imported goods not included.)



Source: S. Sibley and G. Matos, U.S. Geological Survey

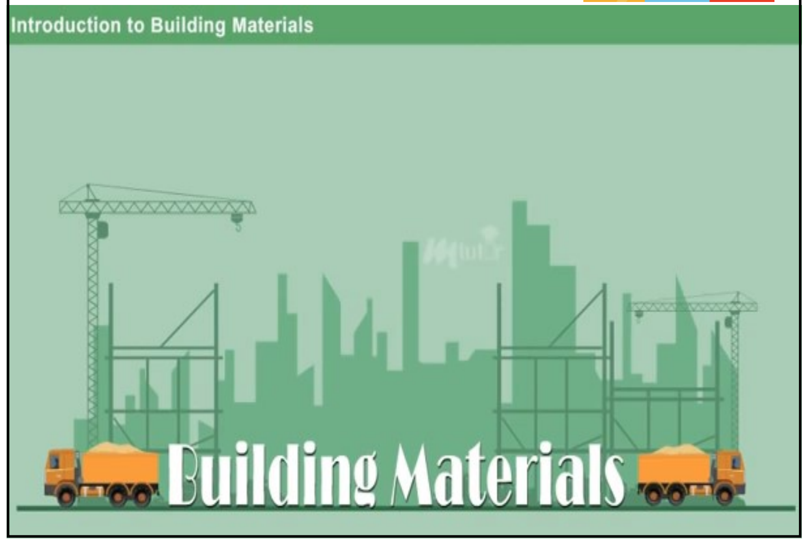
Course Objective

- The course deals with properties, characterization, testing, selection and quality control of construction material.
- For economical, durable and high performance of structure, it is imperative to have knowledge on construction materials and its wider applications.
- This course provides the basic and enhanced overview on various construction materials presently used in practice.

Course Structure

❖ 40 lectures (+ tutorials)
- ✓Cement, Concrete
- ✓Lime
- ✓Masonry
- ✓Wood
- ✓Highway materials
- ✓Steel
- ✓Polymeric material and Geo-synthetics

❖ Instructors will discuss selected topics and questions during tutorials



Course Structure

❖ 40 lectures (+ tutorials)
- ✓Cement, Concrete
- ✓Lime
- ✓Masonry
- ✓Wood
- ✓Highway materials
- ✓Steel
- ✓Polymeric material and Geo-synthetics

❖ Instructors will discuss selected topics and questions during tutorials

Syllabus

No. of Lectures	Learning Objective
6	Overview, Concrete as a construction material, cement and hydration
4	Aggregate, chemical admixtures
5	Concrete mix design, fresh concrete,
6	Hardened concrete properties, testing of concrete
4	Durability of concrete and special concretes
4	Clay products, lime: Properties, tests, selection
3	Wood and timber
3	Steel, ferrous and non-ferrous metals
5	Highway materials, Polymeric material, geo-synthetics, civil engg equipments
Total = 40	

Examination and grading

- Evaluated tutorials (15%)
  - **Best 5 out of 6 (no make-ups)**
- Mid-semester Exam (25%)
- Lab Experiments (20%)
  - Lab record and viva (10%), Lab quiz (10%)
- Comprehensive exam (40%)

Text books

1. Gambhir, M.L. (2013) "Concrete Technology" Tata McGraw-Hill Publishing Company Ltd, 5th edition.
2. Duggal, S.K. (2019) "Building Materials" New Age International Pvt. Ltd., 5th edition.
3. Gambhir, ML and Jamwal, N. (2014) "Lab Manual- Building and Construction Materials (Testing and quality control)", Mc Graw Hill, 1st edition **(Lab Manual)**

Reference Books:

1. Relevant B.I.S, IRC, ASTM codes
2. Mehta, P.K. and P.J.M. Monteiro. (2006) "Concrete Microstructure, Properties, and Materials", The McGraw-Hill Companies, United States.
3. Neville, A.M. (1995) "Properties of Concrete", Pearson Education, 4th edition
4. Gambhir, M.L., and Jamwal, N. (2011) "Building Materials- Products, Properties and Systems", Mc Graw Hill, 1st edition
5. Moondra, HS and Gupta, R. (2000) "Lab Manual for Civil Engineering", CBS, 2nd edition **(Lab Manual)**